Q1.

a) 120 = 01111000
+62 = 00111110

1111
01111000
+ 00111110
10110110 ≠

~~100~~ 10110110
2's cmp 01001010 = 74
~~-74~~ So, decimal = -74

120+62 = 182 . but it is outside (-128 to 12
range.
So overflow occured

b) (-120) + (-62)
-120 => 10000111
+ 1
10001000

-62 => 00111110
1's cmp 11000001
+ 1
11000010

10001000
+ 11000010
$\cancel{1}$ 01001010 ⇒ 74.

2) a) Left shift

10110110 —— 10110110

10110000

Ans: 10110000 ⇒ 176

b) 11010100

00011010 ⇒ 26